LERAY ACYCLICITY VIA THE STUPID TRUNCATION FILTRATION

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Allowed standard prerequisites (editorial).

- Two-out-of-three for defined cohomology. lemma-2-out-of-3-defined (same chapter, carded in the old generation). Engine applied at each truncation stage of the induction.
- Negative vanishing. lemma-negative-vanishing (same chapter, carded in the old generation). Closes the identification at the end of the argument.
- Derived functor setup and truncation sections. definition-derived-functor, homology-section-truncations, lemma-2-out-of-3-computes (same chapter / cross-chapter, cited in-span; not carded). Supply the section maps of the stupid truncation pieces and the computes-side of the two-out-of-three engine.

Source reference. Original definition 0157: sources/stacks-derived/pinned-original-source.tex, source lines 5495–5495. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0118: focused-seeds/geometry/S9-der-leray-acyclicity/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SZ: sources/stacks-derived/pinned-original-source.tex, source lines 5234–5234. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SC: sources/stacks-derived/pinned-original-source.tex, source lines 4896–4896. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TC: sources/stacks-derived/pinned-original-source.tex, source lines 5714–5714. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (Leray's acyclicity lemma)**.** *Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories. Let $A^\bullet$ be a bounded below complex of right F-acyclic objects such that RF is defined at $A^\bullet$[1]. The canonical map*

$$F(A^\bullet) \longrightarrow RF(A^\bullet)$$

is an isomorphism in $D^+(\mathcal{B})$, i.e., $A^\bullet$ computes RF.

Proof. Let $A^\bullet$ be a bounded complex of right F-acyclic objects. We claim that RF is defined at $A^\bullet$ and that $F(A^\bullet) \to RF(A^\bullet)$ is an isomorphism in $D^+(\mathcal{B})$. Namely, it holds for complexes with at most one nonzero right F-acyclic object by Definition 0157. Next, suppose that $A^n = 0$ for $n \notin [a, b]$. Using the "stupid" truncations we obtain a termwise split short exact sequence of complexes

$$0 \to \sigma_{\geq a+1} A^\bullet \to A^\bullet \to \sigma_{\leq a} A^\bullet \to 0$$

see Homology, Section 0118. Thus a distinguished triangle $(\sigma_{\geq a+1} A^\bullet, A^\bullet, \sigma_{\leq a} A^\bullet)$. By induction hypothesis RF is defined for the two outer complexes and these complexes compute RF. Then the same is true for the middle one by Lemma 05SZ.

Suppose that $A^\bullet$ is a bounded below complex of acyclic objects such that RF is defined at $A^\bullet$. To show that $F(A^\bullet) \to RF(A^\bullet)$ is an isomorphism in $D^+(\mathcal{B})$ it suffices to show that $H^i(F(A^\bullet)) \to H^i(RF(A^\bullet))$ is an isomorphism for all i. Pick i. Consider the termwise split short exact sequence of complexes

$$0 \to \sigma_{\geq i+2} A^\bullet \to A^\bullet \to \sigma_{\leq i+1} A^\bullet \to 0.$$

Note that this induces a termwise split short exact sequence

$$0 \to \sigma_{\geq i+2} F(A^\bullet) \to F(A^\bullet) \to \sigma_{\leq i+1} F(A^\bullet) \to 0.$$

Hence we get distinguished triangles

$$(\sigma_{\geq i+2} A^\bullet, A^\bullet, \sigma_{\leq i+1} A^\bullet) \quad \text{and} \quad (\sigma_{\geq i+2} F(A^\bullet), F(A^\bullet), \sigma_{\leq i+1} F(A^\bullet))$$

Since RF is defined at $A^\bullet$ (by assumption) and at $\sigma_{\leq i+1} A^\bullet$ (by the first paragraph) we see that RF is defined at $\sigma_{\geq i+2} A^\bullet$ and we get a distinguished triangle

$$(RF(\sigma_{\geq i+2} A^\bullet), RF(A^\bullet), RF(\sigma_{\leq i+1} A^\bullet))$$

See Lemma 05SC. Using these distinguished triangles we obtain a map of exact sequences

$$\begin{array}{ccccccc}
H^i(\sigma_{\geq i+2} F(A^\bullet)) & \longrightarrow & H^i(F(A^\bullet)) & \longrightarrow & H^i(\sigma_{\leq i+1} F(A^\bullet)) & \longrightarrow & H^{i+1}(\sigma_{\geq i+2} F(A^\bullet)) \\
\downarrow & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow \\
H^i(RF(\sigma_{\geq i+2} A^\bullet)) & \longrightarrow & H^i(RF(A^\bullet)) & \longrightarrow & H^i(RF(\sigma_{\leq i+1} A^\bullet)) & \longrightarrow & H^{i+1}(RF(\sigma_{\geq i+2} A^\bullet))
\end{array}$$

By the results of the first paragraph the map β is an isomorphism. By inspection the objects on the upper left and the upper right are zero. Hence to finish the proof it suffices to show that $H^i(RF(\sigma_{\geq i+2} A^\bullet)) = 0$ and $H^{i+1}(RF(\sigma_{\geq i+2} A^\bullet)) = 0$. This follows immediately from Lemma 05TC. □

[1]For example this holds if $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ is everywhere defined.